

COP2800C Module 4 Graded Programming Assignment

I recommend that you use the Horizon system for this assignment unless you are comfortable working with environment variables and execution search paths. All necessary configurations are already in place on the Horizon system. The Module 3 solution has been included in the assignment repository for your convenience, or you can use your own solution if any necessary corrections have been made based on my grading comments.

Make the following changes to the Penguin Rookery from the Module 3 solution:

- Split the classes and enum type into separate files. Declare the classes and enum as public. Remember that the file names must *exactly* match the class/enum names.
- Add a package statement to the top of each file (after the ID header) and before any other declarations. Add a blank line before and after the package statement. Use the following package specification: edu.fscj.cop2800c.penguins
- Use jGrasp to save your files in the proper location as described in the practice exercise.
- Build the program by compiling the PenguinRookery file **from the command line** as demonstrated in the practice exercise. The compiler will find the other files as necessary, assuming you have located them in the correct folder.
- Execute your program by executing the PenguinRookery class **from the command line** as demonstrated in the practice exercise.

Attach and submit the following artifacts (a fancy name we use in software development for documents used/produced by a program):

- Your **three** source code files
- One or more screen snips which show **successful compilation and execution (both!) from the command line** as demonstrated in the practice exercise. Do not snip source code, do not snip the jGrasp IDE image; only snip the compile operation and the execution output.

Expected output is as follows (has not changed from Module 3):

```
Our rookery contains the following penguins:
First penguin is a(n) CHINSTRAP.
Second penguin is a(n) GENTOO.
And the final penguin is a ADELIE.
There are a total of 3 penguin species in our rookery.
Testing mutators and accessors
first penguin: ADELIE
second penguin: CHINSTRAP
third penguin: GENTOO
```